Lesson 2.6 – 2.9: Quadratic Functions Test Outline

[Lesson 2.6 – 2.7 & 2.9] Using the discriminant

	Quadratic Relation	Quadratic Equation	Linear – Quadratic System
Standard Form	$y = ax^2 + bx + c$	$0 = ax^2 + bx + c$	$mx + k = ax^2 + bx + c$
The discriminant is used for?	Find the number of zeros / x-intercepts	Find the nature of the roots	Find the Points of Intersection (POIs)
$b^2 - 4ac > 0$ when a and c have opposite signs	2 x-intercepts / zeros	2 distinct real roots	2 POIs
$b^2 - 4ac = 0$ When perfect square trinomial	1 x-intercept / zero	2 equal real roots	1 POI
$b^2 - 4ac < 0$	No x-intercepts / zeros	2 imaginary roots	No POIs

Using an Interval Table for Quadratic Inequalities involving the Discriminant

		Root #1		**Root #2**	
	$x < s$		$s < x < t$		$x > t$
$(x - s)$	−		+		+
$(x - t)$	−		−		+
$a(x - s)(x - t)$	+		−		+

(switch all signs if a is negative)

[Lesson 2.8] Determining the algebraic model of a family of quadratic relations

- Family of quadratic relations all have the same zeros and the same axis of symmetry.
  - Substitute zeroes into $y = a\,(x - s)(x - t)$
  - Substitute the given third point (e.g. y-intercept, any point) for x, y to determine value of a.

[Lesson 2.9] Solving the linear-quadratic system of equations (finding the POIs)

- Equate the linear and quadratic (i.e. $y_1 = y_2$) and simplify into a new quadratic equation.
- Solve the equation to find the x-coordinates of the POIs, then use y_1 or y_2 to find the y-coordinates.

Practice Questions

Lesson 2.6

1. How many zeros do these relations have?
   a) $y = x^2 + 7x + 3$ b) $y = -3x^2 - 4x - 10$ c) $y = 2x^2 - 4x + 2$

2. Determine the nature of the roots of the quadratic equation $0 = 4x^2 - 4x + 1$.

3. Describe why the discriminant is useful in finding how many x-intercepts a quadratic relation has.

4. Explain the difference between finding the number of zeros and finding the nature of the roots.

Lesson 2.7

5. Find the value(s) of k such that $kx - 1 = 2x^2$ will have two equal real roots.

6. Find the value of k such that $3x^2 + 4x + k = 0$ has:
   a) two equal root b) two distinct roots c) two imaginary roots

7. Find the value(s) of k such that $kx - 8 = 2x^2$ has:
   a) two equal root b) two distinct roots c) two imaginary roots

8. Without solving, explain why the equation $-3x^2 + kx = -2$ will always have two distinct real roots for all real values of k using two different methods.

Lesson 2.8

9. Determine the family of quadratic relation that has zeros $-3 \pm \sqrt{2}$.

10. Determine the quadratic relation, in factored form, that has zeros -3 and 5 and has a y-intercept of -4.

11. Determine the quadratic relation, in standard form, that has zeros $5 \pm \sqrt{3}$ and has a minimum value of -20.

12. Find $y = ax^2 + bx + c$ such that it passes through the points $(0, -1), (5, -11), (-1, -5)$.

Lesson 2.9

13. Determine the number of solutions to the linear-quadratic system without solving.

$$y = x - 2$$

$$y = x^2 - 5x + 7$$

14. Solve the following linear-quadratic system.
   a) $y = -x^2 - 3x - 5$, $y = 2x + 1$
   b) $y = x^2 - 5x + 7$, $y = x - 2$
   c) $y = -x^2 - 3x - 5$, $x + y = 9$

15. Determine the value(s) of k so that the system below will have exactly one solution.
$y = 2x^2 - 5x + 3, \quad y = 3x + k$

16. Determine the value of the y-intercept of a line with a slope of -2 that is a tangent line to $y = 3x^2 - 4x + 1$.

17. Find the value(s) of k such that $y = -8x^2 + 4x + k$ if it intersect with the line $y = 2x - 2$ at:
a) one point b) two points c) no points

18. Describe one similarity and one difference between the methods of 1) finding the zeros of a quadratic and 2) finding the point(s) of intersection between a parabola and a line.

Answers:

1a) 2 zeros b) no zeros c) 1 zero
2) 2 equal real roots
3) The quadratic formula contains the radicand, $b^2 - 4ac$ (the discriminant). If it is less than zero, there will be no solutions because there is no real answer for a square of a negative number. If the radicand is 0, then there is one x-intercept, since the square root of 0 is 0. If the radicand is greater than 0, then there are two x-intercepts because x can be either positive or negative.
4) A quadratic relation is a relation between 2 variables and you are finding the x value(s) of the relation when y = 0 which are zeros/x-intercepts. You can have 2, 1, or no zeros. A quadratic equation is a polynomial with a degree of 2 that must always have 2 real solutions to make LS = RS and the nature of the roots tells the type of solutions: two distinct real roots, two equal real roots or two imaginary roots.
5) $\pm 2\sqrt{2}$
6a) $k = \frac{4}{3}$ b) $k < \frac{4}{3}$ c) $k > \frac{4}{3}$
7a) $k = \pm 8$ b) $k > 8 \; or \; k < -8$ c) $-8 < k < 8$
8) method #1 - a and c are opposite signs (parabola opens down and has a positive y-intercept); method #2 – the discriminant $k^2 + 24$ is always positive since k squared is positive and you are adding 24
9) $y = a(x^2 + 6x + 7)$
10) $y = \frac{4}{15}(x + 3)(x - 5)$
11) $y = \frac{20}{3}x^2 + \frac{200}{3}x + \frac{440}{3}$
12) $y = -x^2 + 3x - 1$
13) 1 POI
14a) (-2, 3), (-3, -5) b) (3, 1) c) no POIs
15) -5
16) $\frac{2}{3}$
17a) $k = \frac{-17}{8}$ b) $k > \frac{-17}{8}$ c) $k < \frac{-17}{8}$
18) One similarity is that in both we can use one of the following methods: factoring, completing the square (then isolating for x) or using the quadratic formula to find the x-coordinates. One difference is that in 1), we are finding the x-intercepts of the quadratic so we do not need to solve for y (we already know that y = 0); in 2), we first have to find the x-coordinates of the P.O.I.'s, then substitute them into one of the two equations (the linear or the quadratic) to find the y-coordinates of the P.O.I.'s.